

PROJECT REPORT

ATM CASH REPLEMISHMENT

Using PYTHON

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1. Introduction

Automated Teller Machines (ATMs) play an important role in providing customers with 24×7 access to banking services such as cash withdrawals and deposits. One of the major challenges faced by banks is maintaining an adequate amount of cash in ATMs. Traditional cash replenishment methods are often based on fixed schedules or manual estimation, which may lead to cash shortages or excess cash, increasing operational costs and affecting customer satisfaction.

This project, "ATM Cash Replenishment Optimization Using Machine Learning," aims to develop an intelligent system that predicts the amount of cash required for ATM replenishment based on historical transaction data and various influencing factors such as withdrawals, deposits, weekends, holidays, festivals, weather conditions, and nearby events. The prediction model is developed using the Random Forest Regression algorithm, which provides accurate and reliable forecasting.

The application is implemented using Python, CustomTkinter, SQLite, and Scikit-learn. It provides a user-friendly interface with modules for cash prediction, analytics, database management, and prediction history. The proposed system helps banks optimize cash replenishment, reduce operational costs, improve cash availability, and enhance overall ATM management efficiency.

2. Problem Statement

- Traditional ATM cash replenishment is based on fixed schedules or manual estimation.
- Cash demand at ATMs varies due to weekends, holidays, festivals, weather conditions, and local events.
- Incorrect cash estimation can result in **cash shortages**, causing inconvenience to customers.
- Excess cash in ATMs increases operational and maintenance costs for banks.
- Manual cash management is time-consuming and less efficient.
- There is a need for an intelligent system that can accurately predict ATM cash requirements.
- The proposed system uses **Machine Learning (Random Forest Regression)** to forecast the required cash amount based on historical transaction data.
- The system aims to optimize cash replenishment, reduce operational costs, and ensure continuous cash availability at ATMs.

3. Objectives

- To develop an intelligent system for predicting ATM cash replenishment requirements.
- To improve the accuracy of cash demand prediction using Machine Learning.
- To reduce unnecessary cash replenishment trips and operational costs.
- To minimize ATM cash shortages and ensure continuous cash availability.
- To provide a user-friendly desktop application for ATM cash management.
- To maintain prediction history using a database for future analysis.
- To generate analytical reports and visualize transaction trends.

4. Scope of the Project

- Predict ATM cash requirements using historical transaction data.
- Analyze factors such as weekends, holidays, festivals, weather, and nearby events.
- Provide an easy-to-use desktop application for bank staff.
- Store prediction records in a database for future reference.
- Display analytical graphs for better decision-making.
- Improve ATM cash management and operational efficiency.
- The system can be further extended for real-time banking applications

5. Tools & Technologies Used

The development of the **ATM Cash Replenishment Optimization Using Machine Learning** project involved the use of various software tools, programming libraries, and technologies. These technologies were selected to ensure efficient data processing, accurate prediction, and an interactive user interface.

Python

Python was used as the primary programming language for developing the application. It provides extensive support for data analysis, machine learning, database connectivity, and graphical user interface development.

CustomTkinter

CustomTkinter was used to design a modern and user-friendly Graphical User Interface (GUI). It provides attractive widgets and improves the overall user experience compared to the standard Tkinter library.

Pandas

Pandas was used for loading, cleaning, preprocessing, and analyzing the ATM transaction dataset. It simplifies handling large datasets and performing various data manipulation tasks.

NumPy

NumPy was used for numerical computations and efficient manipulation of arrays and matrices required during data preprocessing and model training.

Scikit-learn

Scikit-learn is the machine learning library used to train the **Random Forest Regression** model. It provides algorithms for data splitting, model training, prediction, and performance evaluation.

Matplotlib

Matplotlib was used to generate analytical graphs and charts, such as monthly withdrawals, holiday analysis, weekend analysis, festival analysis, and city-wise cash withdrawal trends.

SQLite

SQLite was used as the database management system to store prediction history and other application data. It is lightweight, easy to use, and suitable for desktop applications.

Joblib

Joblib was used to save and load the trained machine learning model (atm_model.pkl). This eliminates the need to retrain the model every time the application starts.

Visual Studio Code (VS Code)

Visual Studio Code served as the Integrated Development Environment (IDE) for writing, debugging, and managing the project's source code.

Git and GitHub

Git was used for version control, while GitHub was used to store the project repository and manage the source code efficiently.

6. System Requirements

Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 500 MB of free disk space
- Operating System: Windows 10/11
- Display Resolution: 1366 × 768 or higher

Software Requirements

- Programming Language: Python 3.x
- IDE: Visual Studio Code (VS Code)
- Machine Learning Library: Scikit-learn
- GUI Framework: CustomTkinter
- Database: SQLite
- Data Analysis Libraries: Pandas and NumPy
- Visualization Library: Matplotlib
- Model Serialization: Joblib
- Version Control: Git & GitHub

Functional Requirements

- Predict ATM cash replenishment using historical data.
- Accept user inputs such as withdrawals, deposits, holidays, weekends, festivals, weather conditions, and nearby events.
- Display the predicted cash replenishment amount.
- Store prediction history in the database.
- Display analytical graphs and reports.
- Provide a user-friendly graphical interface.
- Allow users to manage prediction records.

Non-Functional Requirements

- Provide accurate and reliable predictions.
- Ensure fast response time for prediction generation.
- Maintain data integrity and security.
- Support easy maintenance and future enhancements.
- Provide an intuitive and responsive user interface.
- Ensure compatibility with Windows operating systems.

7. System Architecture

Overview

The **ATM Cash Replenishment Optimization System** is designed using a modular architecture that integrates data processing, machine learning, database management, and a graphical user interface. The system collects ATM-related data from the user, processes it using a trained **Random Forest Regression** model, predicts the required cash replenishment amount, and stores the prediction in a SQLite database. The application also provides analytical graphs and prediction history through an interactive dashboard.

Working of the System

- The user enters ATM-related information through the Prediction module.
- The input data is preprocessed and converted into the required format.
- The trained **Random Forest Regression** model predicts the cash required for replenishment.
- The prediction result is displayed to the user.
- The prediction is stored in the **SQLite** database.
- The Analytics module displays graphs for transaction analysis.
- The Dashboard provides an overview of the system and quick navigation to all modules.

8. Methodology

The development of the **ATM Cash Replenishment Optimization Using Machine Learning** system followed a systematic approach to ensure accurate prediction and efficient cash management. The methodology adopted for this project is described below:

➤ Data Collection

Historical ATM transaction data was collected, including details such as opening cash, withdrawals, deposits, weekends, holidays, festivals, weather conditions, nearby events, city, month, and day. This data forms the basis for training the machine learning model.

➤ Data Preprocessing

The collected dataset was cleaned to remove missing or inconsistent values. Categorical features such as city names were encoded into numerical values, and the data was prepared for model training.

➤ Feature Selection

Relevant features affecting ATM cash demand were selected. These features include withdrawal amount, deposits, holidays, weekends, festivals, weather conditions, nearby events, and other transaction-related information.

➤ Model Training

The processed dataset was divided into training and testing sets. A **Random Forest Regression** algorithm was used to train the model because of its high prediction accuracy and ability to handle complex datasets.

➤ **Model Evaluation**

The trained model was tested using the testing dataset. Performance was evaluated using metrics such as **Mean Absolute Error (MAE)** and **R² Score** to measure the accuracy and reliability of the predictions.

➤ **Prediction Generation**

The trained model predicts the required cash replenishment amount based on the user-provided ATM information. The prediction helps banks determine the optimal amount of cash required for each ATM.

➤ **Database Storage**

Each prediction is stored in a **SQLite** database along with the input details and prediction result. This enables users to view previous predictions and maintain historical records.

➤ **User Interface**

A modern desktop application was developed using **CustomTkinter**. The application provides modules such as Dashboard, Prediction, Analytics, Database, History, and About, allowing users to interact with the system easily and analyze ATM transaction trends.

9. Dataset Description

The dataset used in this project contains historical ATM transaction records that are used to train and test the Machine Learning model. It includes various factors that influence ATM cash demand, such as customer withdrawals, deposits, holidays, weekends, festivals, weather conditions, and nearby events. The data helps the model identify patterns and accurately predict the amount of cash required for ATM replenishment.

Before training the model, the dataset was preprocessed by removing inconsistencies, handling missing values, and converting categorical data into numerical format. The processed dataset was then divided into training and testing sets to evaluate the performance of the prediction model.

Dataset Features

- **Opening Cash:** Amount of cash available in the ATM at the beginning of the day.
- **Withdrawals:** Total cash withdrawn by customers.
- **Deposits:** Total cash deposited into the ATM.
- **Holiday:** Indicates whether the day is a public holiday.
- **Weekend:** Indicates whether the transaction occurred on a weekend.
- **Festival:** Specifies whether any festival or special occasion occurred on that day.
- **Temperature:** Average daily temperature.
- **Rainfall:** Amount of rainfall recorded on that day.
- **Nearby Event:** Indicates whether any major event took place near the ATM location.
- **City:** Location where the ATM is installed.

- **Month:** Month in which the transaction occurred.
- **Day:** Day of the month.
- **Cash Replenished:** Target value representing the amount of cash to be replenished.

Key Characteristics of the Dataset

- The dataset contains historical ATM transaction records.
- It includes both numerical and categorical attributes.
- Data preprocessing was performed before training the model.
- The dataset was divided into training and testing data for model evaluation.
- **Cash Replenished** was used as the target variable for prediction.

This dataset enables the Random Forest Regression model to learn transaction patterns and generate accurate cash replenishment predictions, helping banks optimize ATM cash management and improve operational efficiency.

10. Machine Learning Model (Random Forest Regression)

Machine Learning plays an important role in predicting the cash required for ATM replenishment. In this project, the **Random Forest Regression** algorithm was used because it provides accurate predictions, handles large datasets efficiently, and reduces the chances of overfitting.

Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees to generate a final prediction. Each decision tree is trained on a different subset of the dataset, and the final output is obtained by averaging the predictions of all the trees. This approach improves prediction accuracy and makes the model more reliable.

The model was trained using historical ATM transaction data after completing the preprocessing stage. Various features such as withdrawals, deposits, weekends, holidays, festivals, weather conditions, nearby events, city, month, and day were used as input to the model. The trained model then predicts the amount of cash that should be replenished in an ATM.

Why Random Forest Regression?

- Provides high prediction accuracy.
- Handles both numerical and categorical data effectively.
- Reduces the risk of overfitting compared to a single decision tree.
- Works well with large datasets.
- Can identify complex relationships between different features.
- Produces stable and reliable predictions.

Working of the Random Forest Model

- The historical ATM transaction dataset is collected.
- The data is cleaned and preprocessed.
- Relevant features are selected for training.
- Multiple decision trees are created using different subsets of the data.
- Each decision tree predicts the cash replenishment amount.
- The predictions of all trees are averaged to generate the final prediction.
- The predicted value is displayed in the application and stored in the database.

Advantages of Using Random Forest

- High accuracy and reliability.
- Less sensitive to noise in the dataset.
- Handles missing values effectively.
- Suitable for complex prediction problems.
- Provides consistent performance on unseen data.
- Easy to train and implement using Scikit-learn.

11. Modules

The **ATM Cash Replenishment Optimization Using Machine Learning** system is divided into several modules. Each module performs a specific function to ensure efficient cash prediction, data management, and user interaction.

➤ Dashboard Module

- Acts as the home page of the application.
- Displays project overview and important statistics.
- Provides quick navigation to all modules.
- Offers a simple and user-friendly interface.

➤ **Prediction Module**

- Allows users to enter ATM-related details.
- Accepts inputs such as opening cash, withdrawals, deposits, holidays, weekends, festivals, weather conditions, nearby events, city, month, and day.
- Uses the trained Random Forest Regression model to predict the required cash replenishment amount.
- Displays the prediction result instantly.

➤ **Analytics Module**

- Displays graphical analysis of ATM transaction data.
- Shows monthly withdrawal trends.
- Provides holiday, weekend, and festival analysis.
- Displays city-wise transaction comparisons.
- Helps users understand cash demand patterns.

➤ **Database Module**

- Stores prediction records in the SQLite database.
- Maintains historical information for future reference.
- Allows efficient storage and retrieval of prediction data.

➤ **Prediction History Module**

- Displays all previously generated predictions.
- Enables users to review historical records.
- Helps in comparing current and past prediction results.

➤ **About Module**

- Provides information about the project.
- Displays the project objectives and technologies used.
- Contains developer details and application information.

➤ Machine Learning Model Module

- Loads the trained Random Forest Regression model.
- Processes user input and generates accurate cash replenishment predictions.
- Integrates the prediction model with the graphical user interface.

12. Results and Output

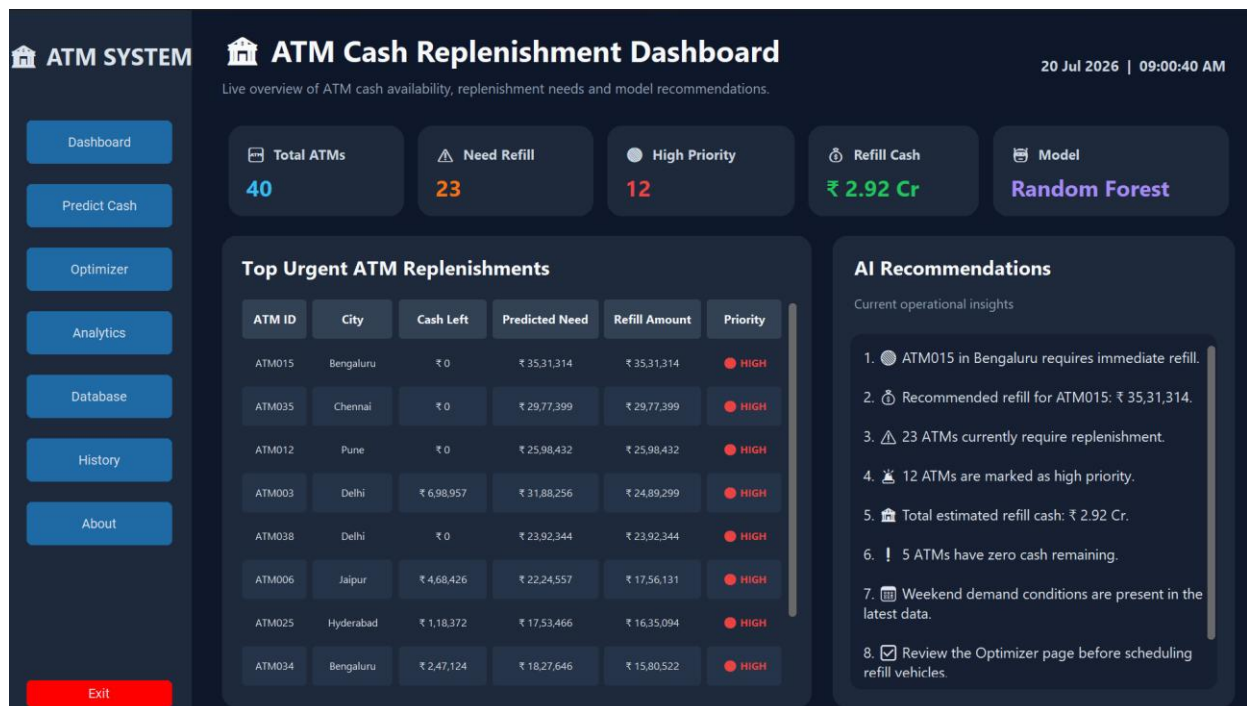
The **ATM Cash Replenishment Optimization Using Machine Learning** system was successfully developed and tested. The application accurately predicts the cash required for ATM replenishment based on user inputs and historical transaction data. The trained **Random Forest Regression** model generates reliable predictions, helping banks optimize cash management.

The graphical user interface developed using **CustomTkinter** provides an easy-to-use environment for users to interact with the system. The Analytics module displays transaction trends through various graphs, while the Prediction History module stores all previous prediction records in the SQLite database.

Output Screens

The application consists of the following screens:

- Dashboard



- Prediction

ATM SYSTEM

Dashboard
Predict Cash
Optimizer
Analytics
Database
History
About
Exit

Cash Prediction

Enter the current ATM conditions to estimate the next-day cash requirement.

ATM Information

Atm Id

ATM001

City

Bengaluru

Location Type

Bus Stand

Day Of Week

Friday

Month

7.0

Is Weekend

0.0

Is Holiday

0.0

Festival

Christmas

Temperature

28.4

Transaction Count

247.0

Average Withdrawal

Withdrawal Amount

Prediction Result

Recommended cash required for the next day

₹ 11,33,243

Risk Level: Low

Current ATM cash position is stable.

Prediction saved to history.

Predict Cash

Reset Fields

- Optimizer

ATM SYSTEM

Dashboard
Predict Cash
Optimizer
Analytics
Database
History
About
Exit

ATM Replenishment Optimizer

Identify ATMs requiring cash replenishment using machine-learning predictions.

Total ATMs

40

Need Refill

23

High Priority

12

Total Refill Cash

₹ 2.92 Cr

All Cities

All Priorities

Apply Filters

Reset

Export Excel

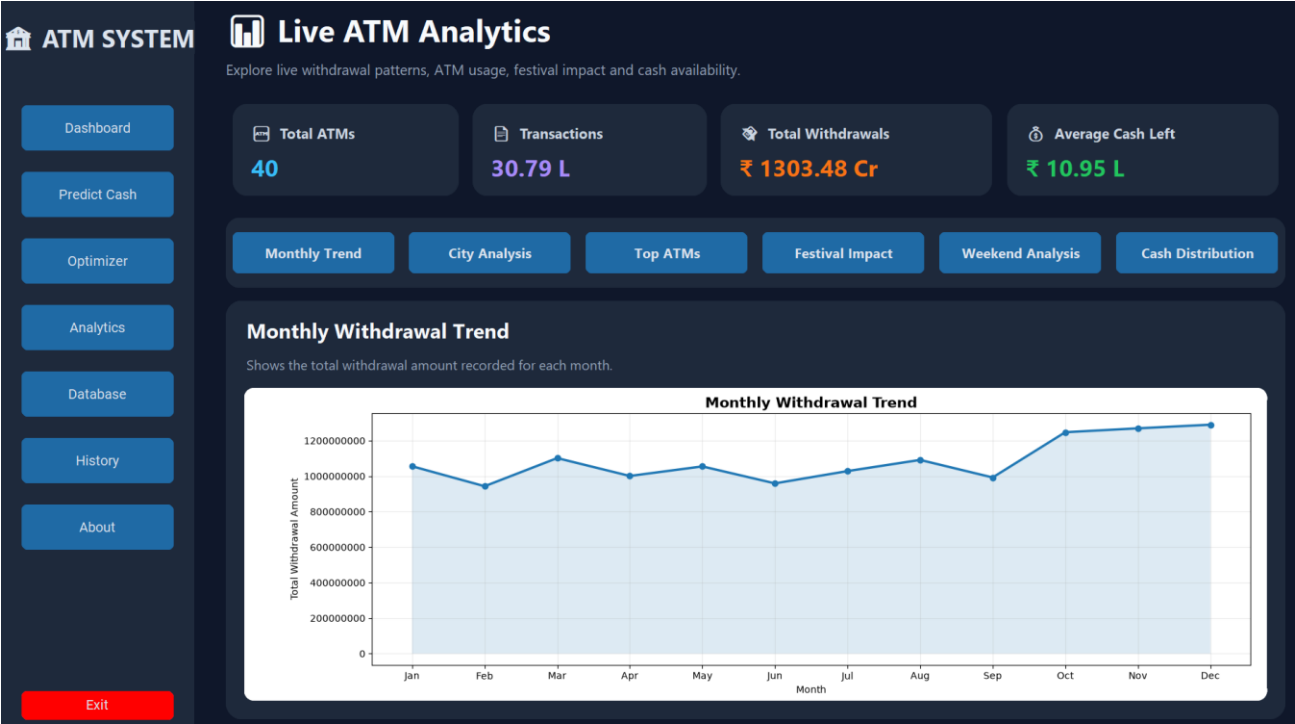
Refresh

ATM Replenishment Priority List

40 records

ATM ID	City	Cash Remaining	Predicted Need	Refill Amount	Priority	Recommended Visit
ATM015	Bengaluru	₹ 0	₹ 35,31,314	₹ 35,31,314	HIGH	Today
ATM035	Chennai	₹ 0	₹ 29,77,399	₹ 29,77,399	HIGH	Today
ATM012	Pune	₹ 0	₹ 25,98,432	₹ 25,98,432	HIGH	Today
ATM003	Delhi	₹ 6,98,957	₹ 31,88,256	₹ 24,89,299	HIGH	Today
ATM038	Delhi	₹ 0	₹ 23,92,344	₹ 23,92,344	HIGH	Today
ATM006	Jaipur	₹ 4,68,426	₹ 22,24,557	₹ 17,56,131	HIGH	Today
ATM025	Hyderabad	₹ 1,18,372	₹ 17,53,466	₹ 16,35,094	HIGH	Today

- Analytics



- Database

ATM SYSTEM

ATM Database Viewer

Database Connected

View, search, export and manage ATM transaction records stored in SQLite.

Total Records

12,000

Unique ATMs

40

Cities

8

Database Size

2.18 MB

All Cities

100

Apply Filters

Reset

Export Excel

Delete Selected

Refresh

ATM Transaction Records

100 records displayed

Selected record: None

ID	ATM ID	Date	City	Location	Transactions	Withdrawal	Opening Cash	Cash Remaining	Cash Out	Refill Required
11518	ATM028	30-12-2025	Pune	Railway Station	262	₹ 14,22,450	₹ 30,00,000	₹ 16,73,089	No	No
11045	ATM022	30-12-2025	Jaipur	Residential	212	₹ 15,92,108	₹ 15,00,000	₹ 1,33,817	No	Yes
9656	ATM028	30-12-2025	Mumbai	College	226	₹ 9,24,987	₹ 25,00,000	₹ 17,16,661	No	No
9625	ATM018	30-12-2025	Delhi	Residential	171	₹ 12,70,092	₹ 25,00,000	₹ 13,72,154	No	No
9004	ATM030	30-12-2025	Bengaluru	Hospital	187	₹ 4,17,603	₹ 30,00,000	₹ 26,15,637	No	No
8515	ATM033	30-12-2025	Hyderabad	College	199	₹ 9,07,413	₹ 30,00,000	₹ 21,42,315	No	No

Exit

- History

ATM SYSTEM

Dashboard

Predict Cash

Optimizer

Analytics

Database

History

About

Exit

Prediction History

Database Connected

Review previous ATM cash predictions, risk levels and operational recommendations.

Total Predictions

27

High Risk

0

Average Prediction

₹ 11.21 L

Unique ATMs

5

All Cite

All Risk

100

Apply Filters

Reset

Export Excel

Delete Selected

Refresh

Saved Prediction Records

27 predictions displayed

Selected prediction: None

ID	Date	ATM ID	City	Location	Opening Cash	Withdrawals	Deposits	Transaction	Prediction	Risk	Recommendation
27	1-07-2026 09:01 A	ATM001	Bengaluru	Bus Stand	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,33,243	LOW	Current ATM cash position is stable
26	1-07-2026 02:55 A	ATM018	Hyderabad	Railway Station	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,86,714	LOW	Current ATM cash position is stable
25	1-07-2026 02:55 A	ATM018	Hyderabad	Mall	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,83,585	LOW	Current ATM cash position is stable
24	1-07-2026 02:55 A	ATM018	Hyderabad	Bus Stand	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,89,000	LOW	Current ATM cash position is stable
23	1-07-2026 02:54 A	ATM018	Hyderabad	Bus Stand	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,88,645	LOW	Current ATM cash position is stable
22	1-07-2026 02:54 A	ATM018	Hyderabad	Bus Stand	₹ 20,00,000	₹ 9,66,746	₹ 90,467	247	₹ 11,86,075	LOW	Current ATM cash position is stable

- About

ATM SYSTEM

Dashboard

Predict Cash

Optimizer

Analytics

Database

History

About

Exit

About ATM Cash Replenishment Optimization

AI Powered ATM Cash Forecasting & Replenishment Management System

Project

Project Name

ATM Cash Replenishment Optimization

Version

1.0

Category

Artificial Intelligence

Prediction Model

Random Forest Regressor

Database

SQLite

Desktop UI

CustomTkinter

Developer

Name

Akshit

Course

Master of Computer Applications

University

Lovely Professional University

Specialization

Data Science

Technologies Used

✓ Python

✓ CustomTkinter

✓ SQLite

✓ Pandas

✓ Scikit-Learn

✓ Matplotlib

✓ Joblib

Features

✓ Cash Prediction

✓ ATM Replenishment Optimizer

✓ Live Analytics Dashboard

✓ SQLite Database

✓ Prediction History

✓ Export to Excel

✓ Machine Learning Integration

Advantages of the System

- Provides accurate cash replenishment predictions.
- Reduces unnecessary cash replenishment trips.
- Minimizes ATM cash shortages.
- Improves operational efficiency.
- Stores prediction history for future analysis.
- Provides graphical reports for better decision-making.
- Easy to use and maintain.

13. Conclusion

The **ATM Cash Replenishment Optimization Using Machine Learning** project successfully demonstrates how machine learning can improve ATM cash management. By using historical transaction data and the **Random Forest Regression** algorithm, the system predicts the amount of cash required for ATM replenishment with good accuracy.

The application provides a modern graphical user interface for entering transaction details, viewing predictions, analyzing graphical reports, and maintaining prediction history. Compared to traditional manual estimation methods, the proposed system offers a more efficient, reliable, and data-driven approach to ATM cash management.

Overall, the project helps reduce operational costs, minimize cash shortages, improve customer satisfaction, and enhance the efficiency of banking operations.

14. Future Scope

The project can be further enhanced by implementing the following features:

- Integration with real-time ATM transaction data.
- Cloud-based deployment for centralized monitoring.
- Mobile application for remote access.
- Automatic cash replenishment alerts.
- Integration with GPS for route optimization of cash vans.
- Use of advanced Deep Learning models for higher prediction accuracy.
- Multi-bank support for centralized ATM management.
- Real-time weather and event API integration for improved forecasting.